

AA6 - Conversion from RE to FA

1

Given the regular expressions $re1$, $re2$ and $re3$, with corresponding automatas $fa1$, $fa2$ and $fa3$, which of the following options describes the correct way to build an automata (following the rules to draw finite automatases) that can recognize the union of the languages defined by $re1$, $re2$ and $re3$?

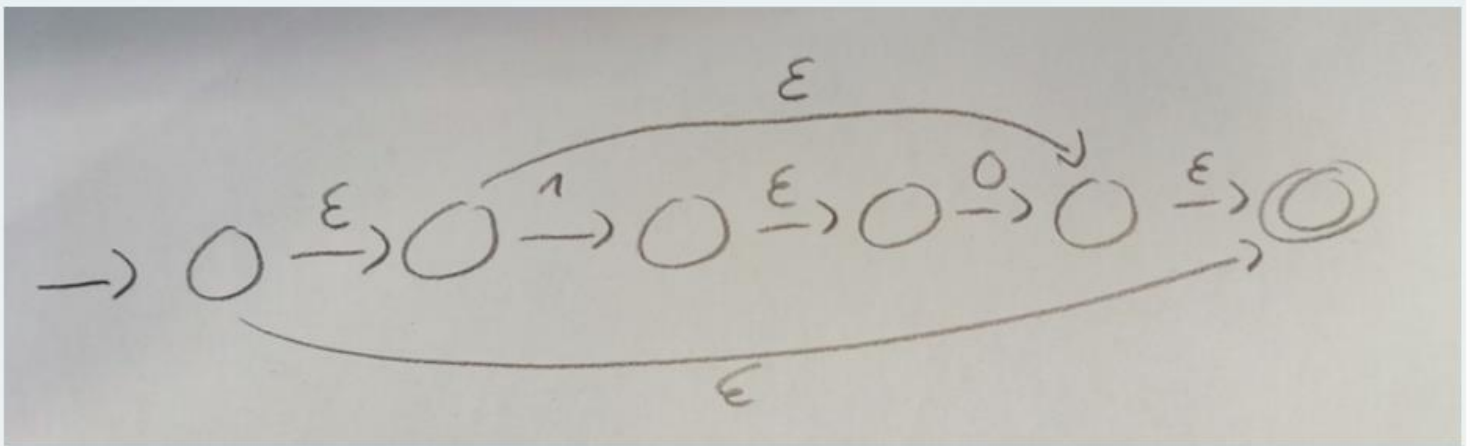
(1/1 Ponto)

- ☐ First we build the automata containing $fa1$ and $fa2$, representing the union of $re1$ with $re2$. Then we add $fa3$ to the automata, representing the union of $re1 + re2$ with $re3$.
- ☐ First we build the automata containing $fa2$ and $fa3$, representing the union of $re2$ with $re3$. Then we add $fa1$ to the automata, representing the union of $re1$ with $re2 + re3$.
- ☐ First we build the automata containing $fa1$ and $fa3$, representing the union of $re1$ with $re3$. Then we add $fa2$ to the automata, representing the union of $re1 + re3$ with $re2$.
- ☐ All of the above.

2

Which of the regular expressions below would originate the following ϵ -NFA (not simplified) by the rules to build a finite automata from a regular expression? 

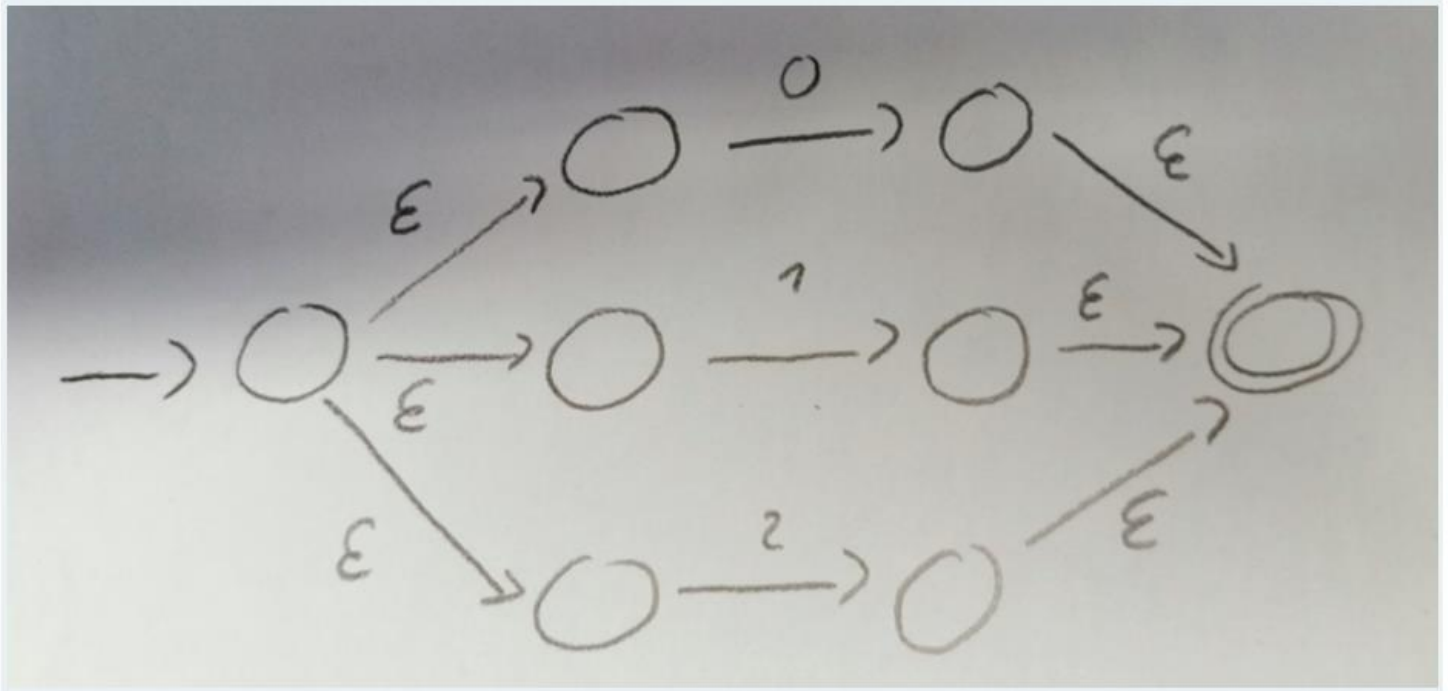
(1/1 Ponto)



- ☐ 10^*
- ☐ $(10)^*$
- ☐ $\epsilon + (10)^*$
- ☐ None of the above.

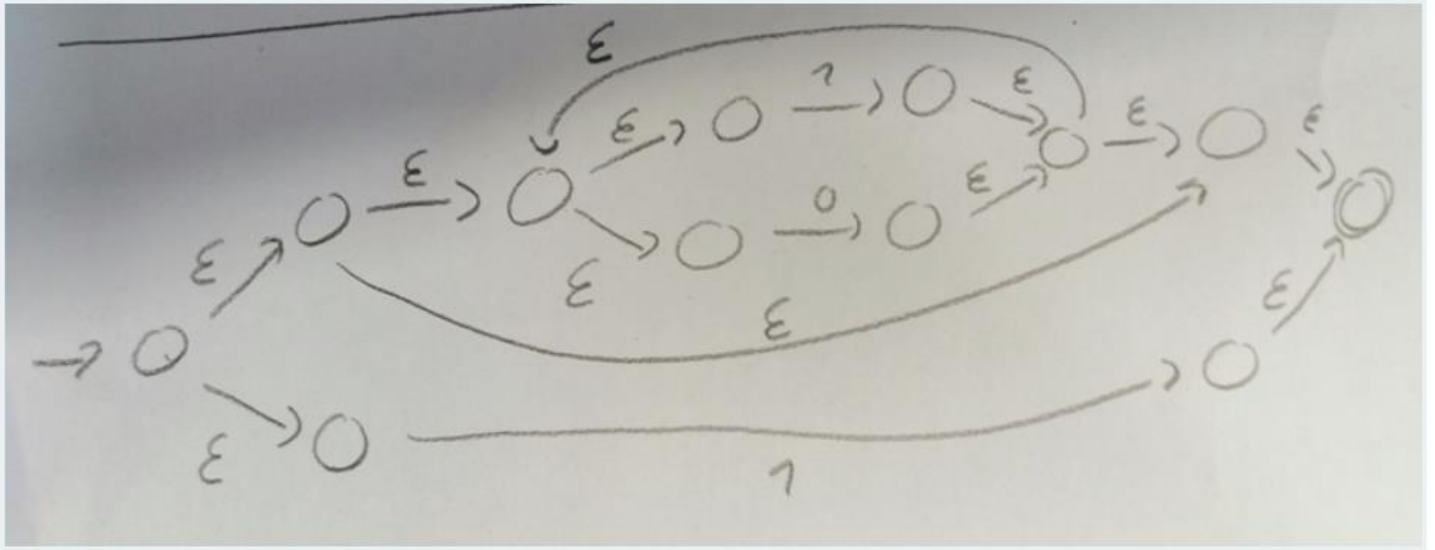
If we strictly apply the rules to build a finite automata from a regular expression, and we do not simplify the result, does the following ϵ -NFA correspond to a correct application of those rules?

(1/1 Ponto)



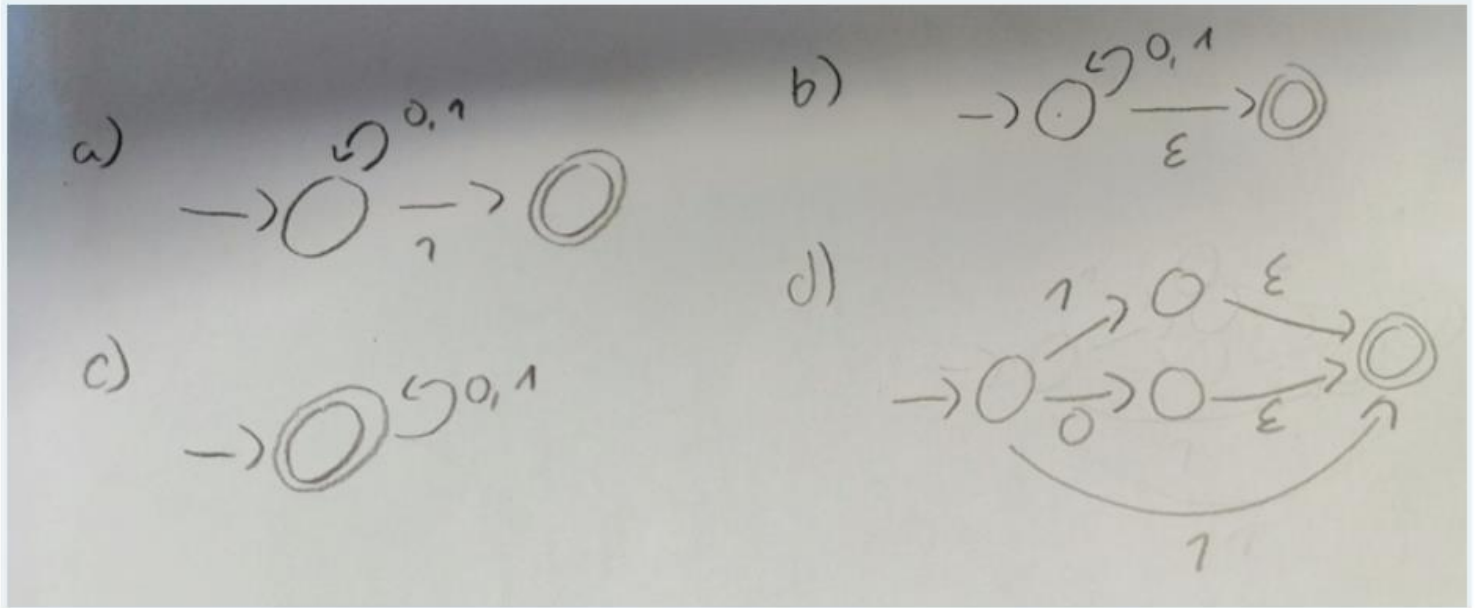
- ☐ Yes, since it corresponds to the union of three finite automata.
- ☐ No, since we would need to first build the union for two finite automatas, and only then add the third.

Which of the following regular expressions generates the following ϵ -NFA (not simplified) when the rules to build a finite automata from a regular expression are applied? 
(1/1 Ponto)



- ☐ $(1 + 0)^* + 1$
- ☐ $(1 + 0)^*$
- ☐ $((1 + 0)^* + \epsilon) + 1$
- ☐ None of the above

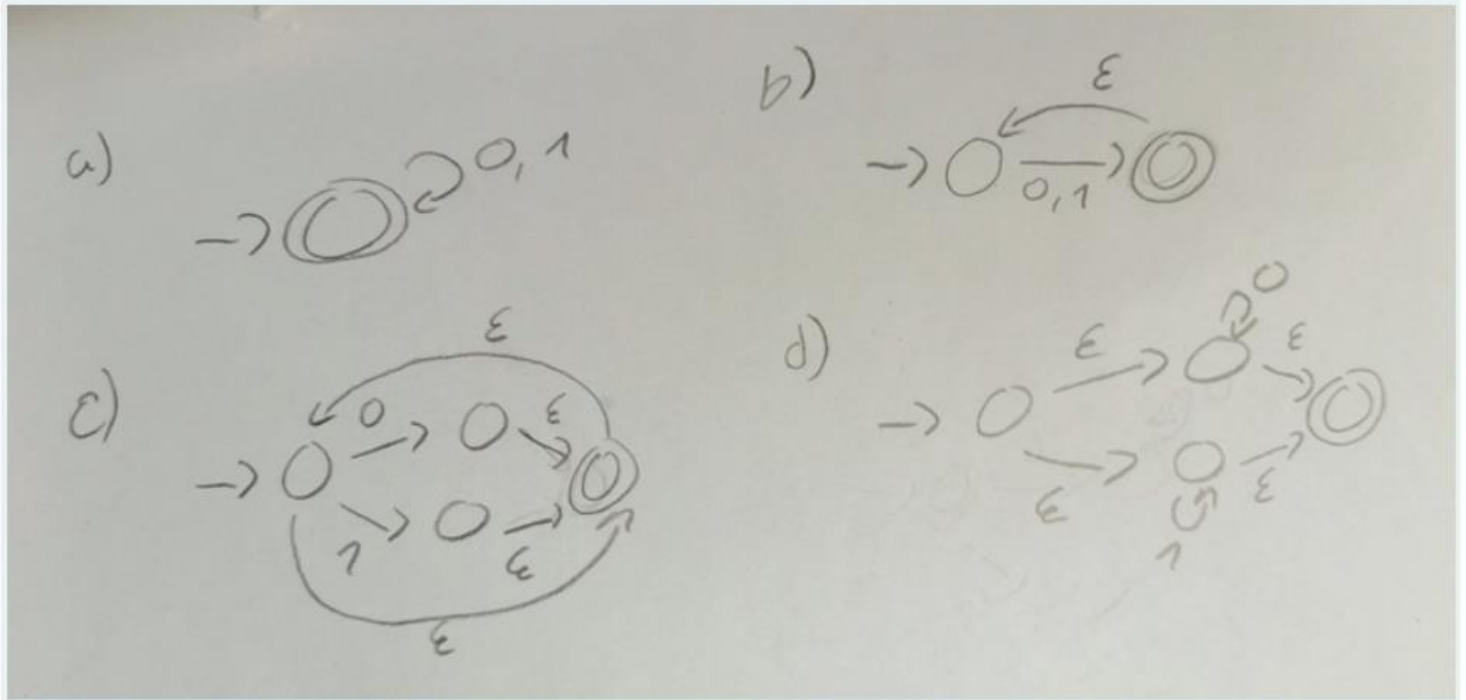
Considering the ϵ -NFA from the previous question, which of the following finite automata would be the result of fully simplifying it?
(2/2 Pontos)



- ☐ FA from option a)
- ☐ FA from option b)
- ☐ FA from option c)
- ☐ FA from option d)
- ☐ None of the above

Which of the following finite automata are equivalent (recognize the same language) among themselves?

(1/1 Ponto)



- ☐ FAs from option a) and b)
- ☐ FAs from option c) and d)
- ☐ FAs from option a) and c)
- ☐ FAs from option b) and d)
- ☐ None are equivalent